



TITLE: Multistage Amplifier

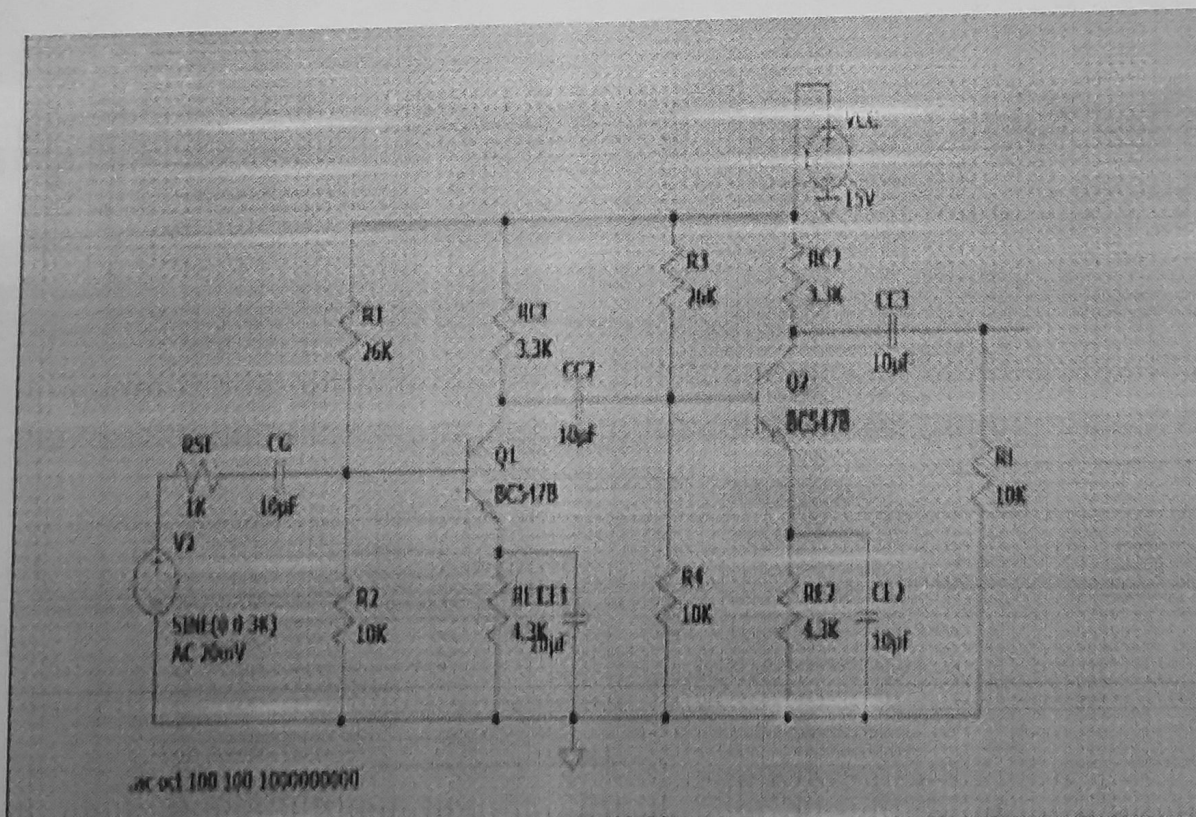
AIM: To implement and study Cascade (CE-CE) and Cascode (CE-CB) structure using BJTs.

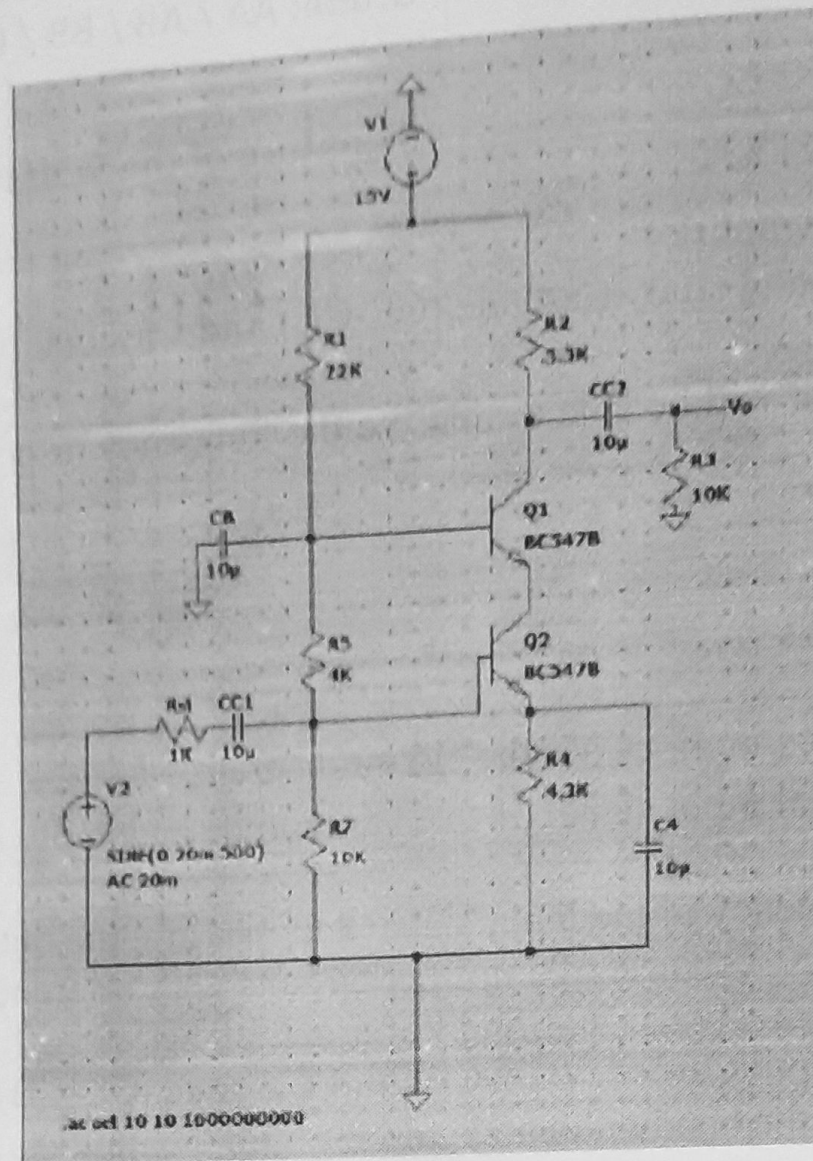
OUTCOME: Student will be able to analyse the frequency response of BJT amplifier.

PROCEDURE:

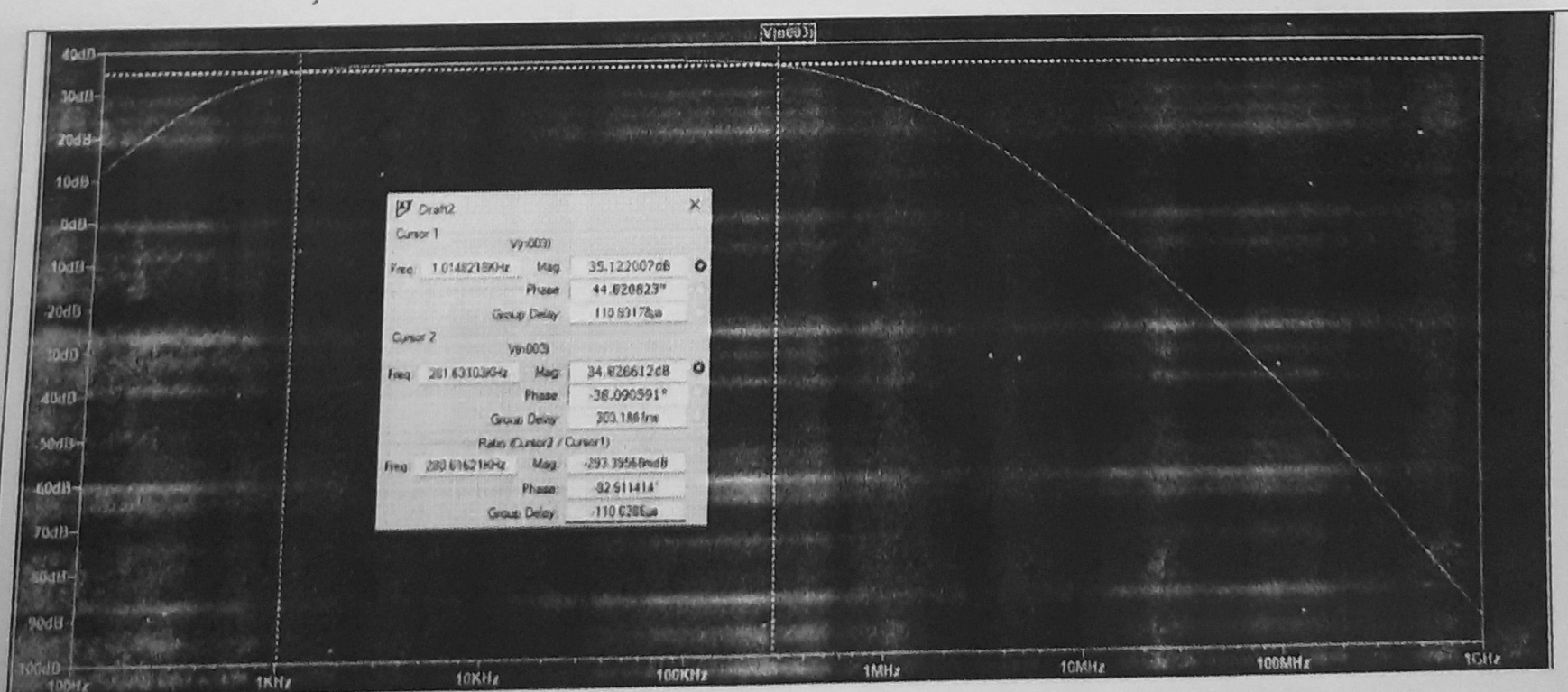
1. Implement the circuit as per the circuit diagram.
2. Take sinusoidal input signal of fixed amplitude and apply it to the input of the amplifier.
3. Vary the frequency of the input signal and measure the output voltage.
4. Find frequency response of amplifier circuits with reference to observation table.
5. Plot frequency response and find lower and higher cut-off frequencies using LTspice for both Cascade (CE-CE) and Cascode (CE-CB) Amplifier
6. Comment on the bandwidth and gain for both the multistage amplifiers

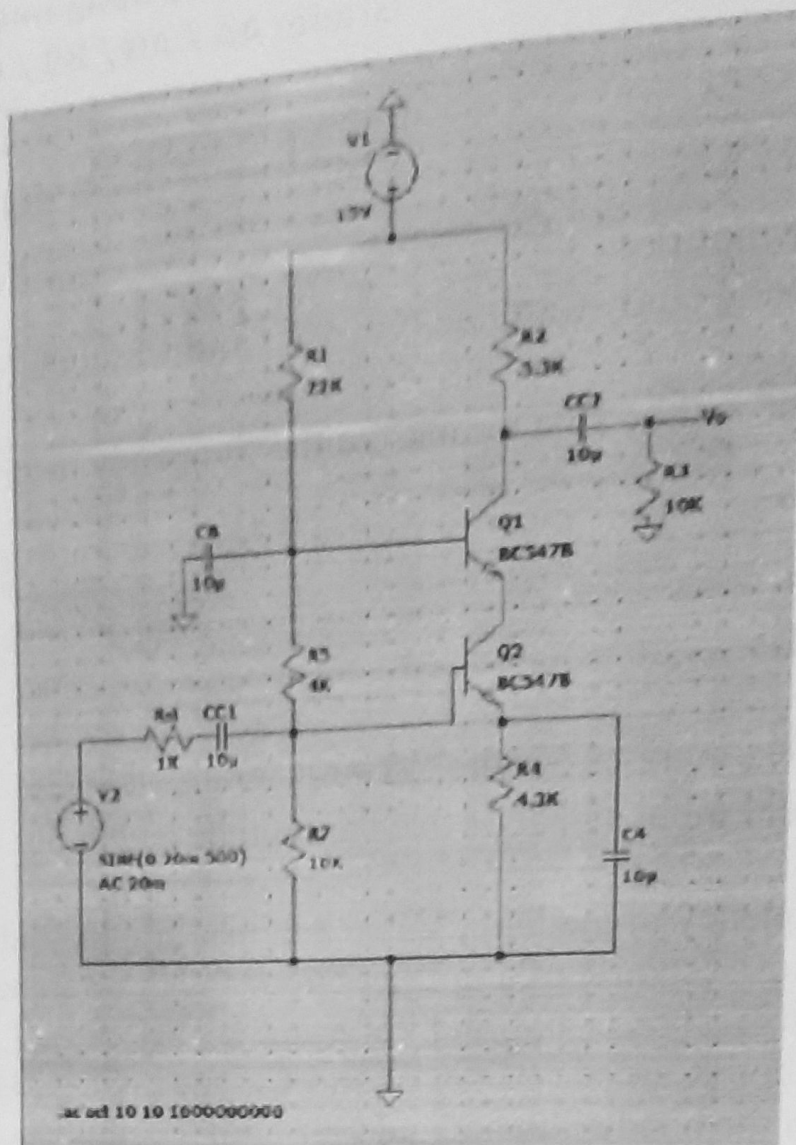
Circuit Diagram:





Observation Table I (Paste the frequency response screenshot of graph from LtSpice)





Observation Table I (Paste the frequency response screenshot of graph from LtSpice)

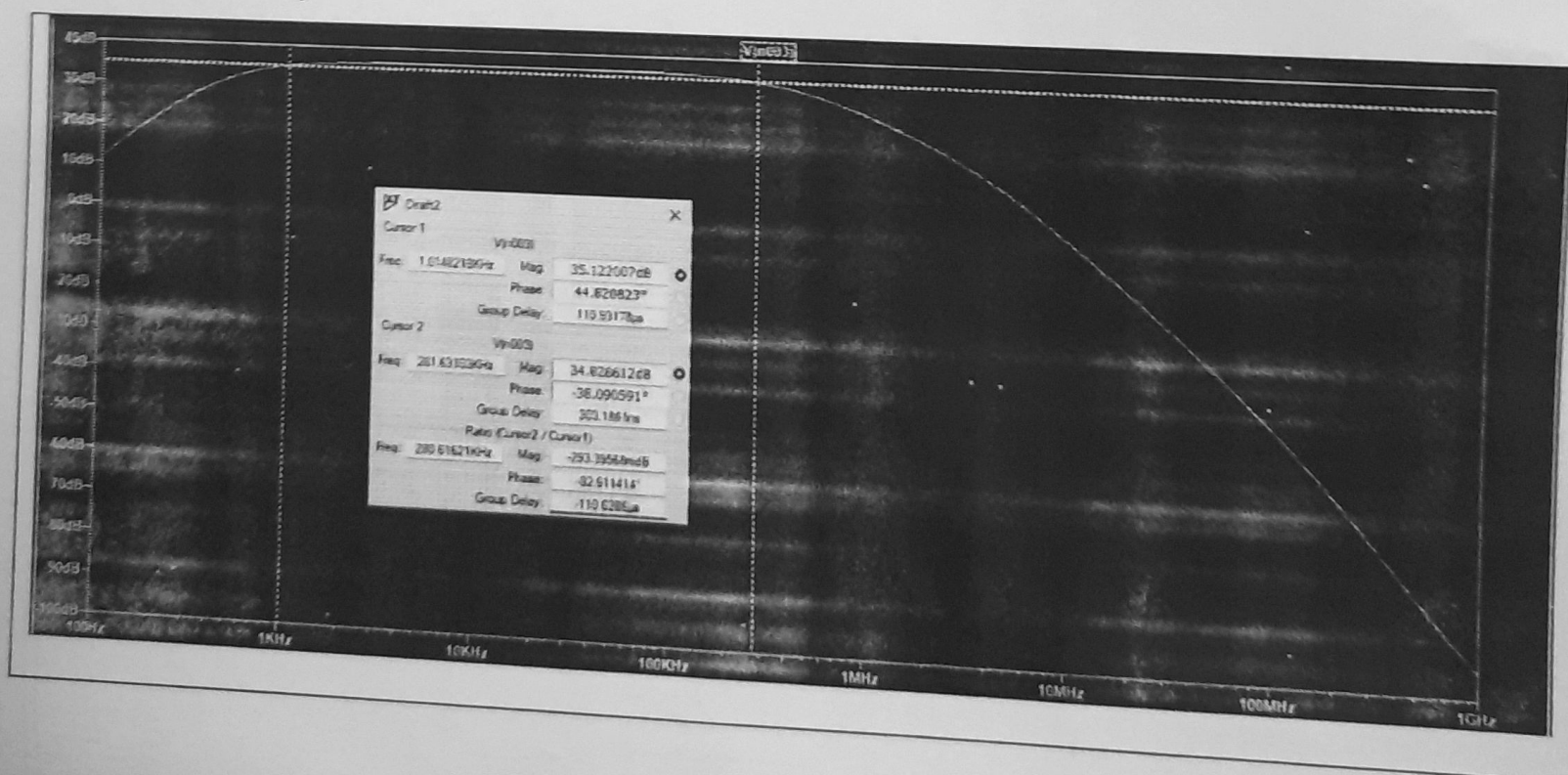
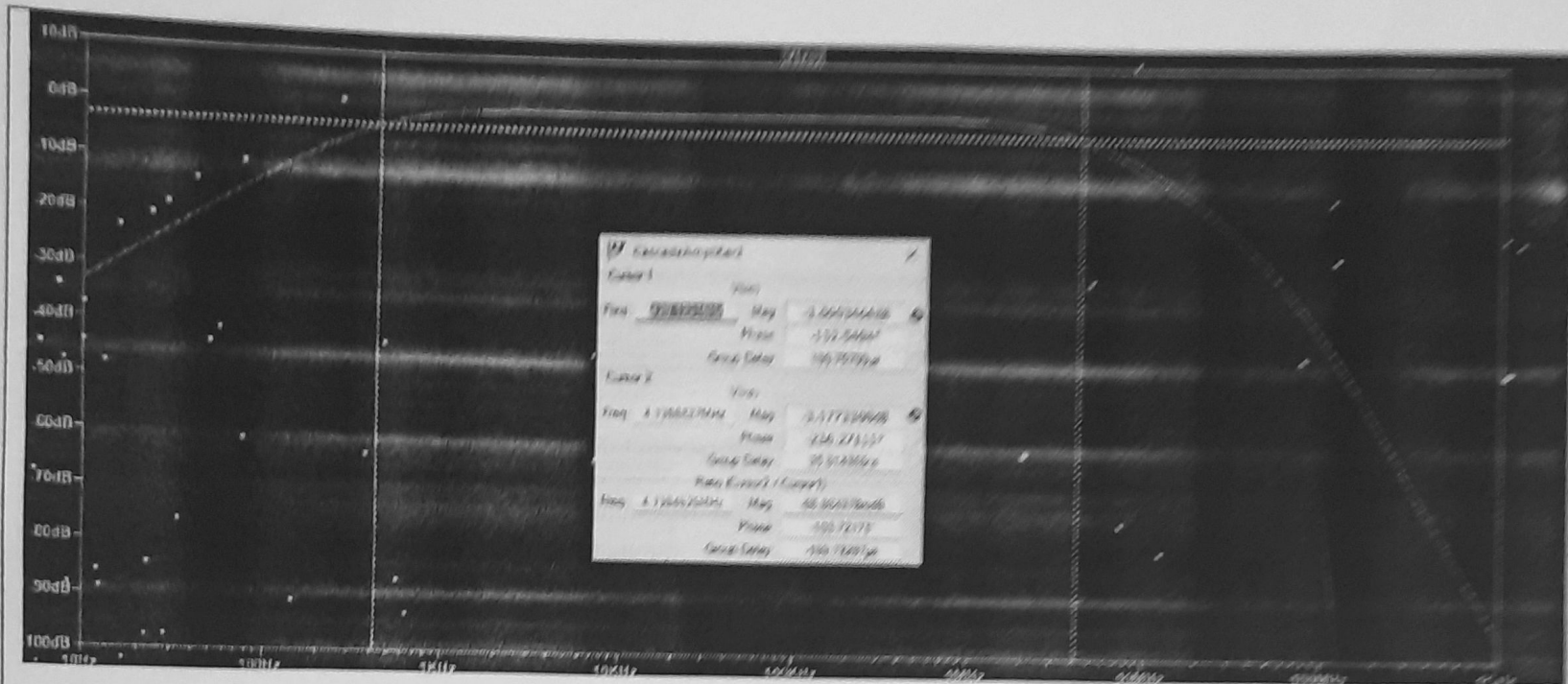


Table II		
Gain (A_v)	3342.5 Practical	4510.2 Theoretical
Lower Cut off Frequency	4.0148 KHz 662 Hz	445 Hz
Higher Cut off Frequency	281.63 KHz 400 kHz	538 kHz
Bandwidth	280.61 KHz 399.4 kHz	537.545 kHz

Observation Table III (Paste the frequency response screenshot of graph from LtSpice)



Observation Table IV

	Practical	Theoretical
Lower Cut off Frequency	420.12 Hz 438.7Hz	459 Hz
Higher Cut off Frequency	4.13888Mhz 3.95 MHz	5.7 MHz
Bandwidth	4.13646 Mhz 3.9572 MHz	5.6 MHz

Gain (A_v)	51.65	76.9
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Signature of faculty in-charge

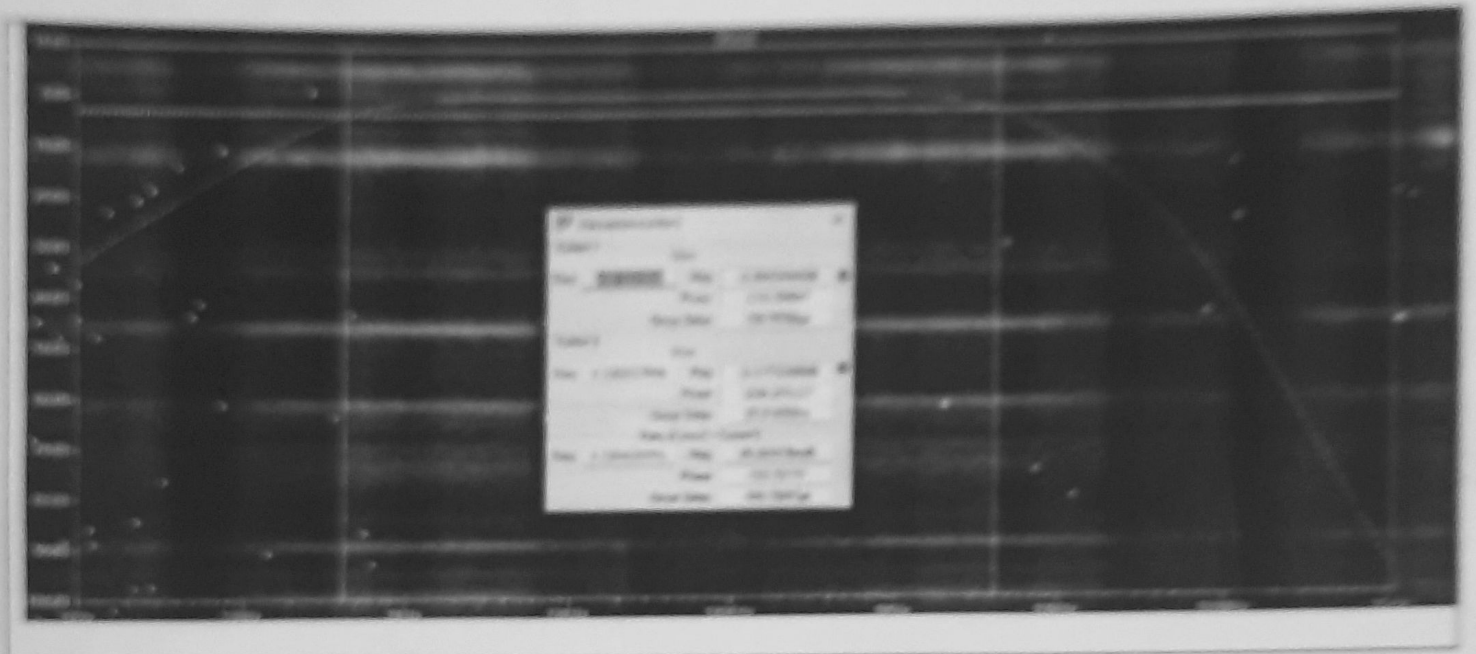


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Observation Table II

Gain (A_v)	3342.5 Practical	4310.2 Theoretical
Lower Cut off Frequency	444.4 Hz 662 Hz	445 Hz
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Observation Table III (Paste the frequency response screenshot of graph from LTSpice)



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Theory:-

Voltage Gain :- Cascade provides extremely high Voltage by multiplying the gains of 2 individual CE stages.

- Cascade: offers moderate gain because the 1st stage acts as a current buffer for 2nd.

Bandwidth :- Cascade: limited by the Miller effect, where the high gain of the 1st stage ~~on~~ magnifies the capacitor, causing it to drop.

Cascade: Designed specifically to combat miller effect by using CB as 2nd stage as ~~2nd stage~~, A_v at unity of first stage causing F_H to drop.

Lower cut-off freq:

- In both Ckt, f_L is determines by CE
- As the biasing and resistor values ($R_E = 4.3k$) are ~~same~~ similar in these specific diagrams, both amplifiers exhibit a similar lower freq

Conclusion:-

- From 'Multistage Amplifier' expt we learned about 2 types of multistage amplifier which are casode & cascade.
- Using LT Spice we observed frequency responses of both the multistage amplifier.
- To achieve good gain we use cascade amplifier and to achieve good ~~gain~~ BW we should use cascode amplifier which is designed to overcome the limitation due to millers effect.

Calculations:

→ Cascade Amplifier:

$$V_{TH} = V_{CC} \frac{R_2}{R_1 + R_2} = 4.17V$$

$$R_{TH} = \frac{R_1 R_2}{R_1 + R_2} = 7.2k\Omega$$

KVL on i/p

$$\therefore -I_B R_{TH} - V_{BE} - (\beta + 1) I_B R_E - \frac{V_{TH}}{\beta} = 0$$

$$\rightarrow I_B = \frac{V_{TH} - V_{BE}}{R_{TH} + (\beta + 1) R_E} = 2.72 \mu A$$

$$\therefore I_C = \beta I_B = 0.801 mA$$

AC Analysis: (MF):

$$g_m = \frac{I_C}{V_T} = 31 mS$$

$$r_{\pi} = \frac{\beta}{g_m} = \frac{300}{31 (10^{-3})} = 9.6k\Omega$$

$$A_{v1} = -g_m (R_{C1} \parallel R_3 \parallel R_4 \parallel R_{TH})$$
$$= -56.1$$

$$A_{v2} = -g_m (R_{C2} \parallel R_5)$$
$$= -76.9$$

$$|A_{v_{mid}}| = |A_{v1}| |A_{v2}| = 4360.2$$

(LF):

$$F_{LCE} = \frac{1}{2\pi R_{eq} C_E} = \frac{1}{2\pi (35)(10)(10^{-6})} = 455 Hz$$

(HF):

$$C_m = C_u (1 + |A_{v1}|)$$
$$= 5 \mu F (1 + 56.7)$$
$$= 288 \mu F$$

$$C_{in} = C_{\pi} + C_m = 308 \mu F$$



$$f_{Hi} = \frac{1}{2\pi R_{eq} C_{in}} = \frac{1}{2\pi (960-28)(308)(10^{-12})} = 538 \text{ Hz}$$

$$\text{Bandwidth} = f_{Hi} - f_{LCC} = 537.5 \text{ kHz}$$

→ Cascode Amplifier

$$V_{th1} = \frac{V_{cc}(R_1)}{R_1 + R_2} = 4.17 \text{ V}$$

KVL on o/p

$$\therefore V_{th1} - V_{BE} - I_{E1} R_E = 0$$

$$\rightarrow I_{E1} = \frac{4.17 - 0.7}{4.3 \text{ k}} = 0.806 \text{ mA}$$

$$\rightarrow I_E = 0.806 \text{ mA}$$

$$\therefore I_C = 0.803 \text{ mA}$$

$$I_B = \frac{0.806(10^{-3})}{301}$$

AC analysis:

$$g_m = 31 \text{ mS}$$

$$r_{\pi} = 9.6 \text{ k}\Omega$$

$$r_e = \frac{1}{g_m} = \frac{1}{31(10^{-3})} = 32.2 \Omega$$

(MF):

$$A_{v1} = -g_m (1/g_m) = -1$$

$$A_{v2} = +g_m (R_C || R_L) = 31(10^{-3})(3.3 || 10) = 76.91$$

$$A_{v \text{ mid}} = -76.9$$

(LF):

$$f_{LCCF} = \frac{1}{2\pi R_{eq} C_{C1}} = \frac{1}{2\pi (34.7)(40 \times 10^{-6})}$$



$$f_{Hi} = \frac{1}{2\pi R_{eq} C_{in}} = \frac{1}{2\pi (960.28)(308)(10^{-12})} = 538 \text{ Hz}$$

$$\text{Bandwidth} = f_{Hi} - f_{LCC} = 537.5 \text{ kHz}$$

→ Cascade Amplifier

$$V_{th1} = \frac{V_{cc}(R_1)}{R_1 + R_2} = 4.17 \text{ V}$$

KVL on o/p

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LF:

$$f_{LCC} = \frac{1}{2\pi R_{eq} C_{in}} = \frac{1}{2\pi (24.7)(10^{-12})}$$

(HF):

$$C_m = C_u (1 + |A_{v1}|) = 10 \mu F$$

$$C_u = C_\pi + C_a \\ = 20 \mu F + 10 \mu F = 30 \mu F$$

$$R_{eq} = R_g \parallel R_1 \parallel R_2 \parallel R_\pi = 1k \parallel 2.85k \parallel 9.6k \\ = \cancel{1k \parallel 2.85k} = 932.6 \Omega$$

$$f_{Hi} = \frac{1}{2\pi (932.6)(10^{-6})(10^{-12})(30)} \\ = 5.68 \text{ MHz}$$

$$B_W = f_H - f_L \\ = f_{Hi} - f_{LCC} = (5.68 \times 10^6 - 458.7) \text{ Hz}$$

$$\rightarrow B_W = 5.6795 \text{ MHz}$$